# Updated Gantry Motor Shortlist

*Quick recommendations for Ricardo and the team | November 3, 2026*

## What changed

**Current prototype:** 3D-printed prototype; 10 lb is an ambitious target, not a proven capacity. Four X/Y motors (two per axis); toothed belts; precision over speed. Attachment motor excluded. Prioritize documented parts and position-error detection.

**Previously discussed:** 50 lb is a future goal, not the initial printed-prototype load. Motor sizing still needs moving mass, pulley sizes and acceleration.

## Three options — ranked for the first prototype

| Rank & system | Price, 4 motors | Pros | Cons / why not first |
|---|---|---|---|
| 1 NEMA 17 closed-loop 0.52 N·m + CL42T-V41 | $195.44 | Smaller package; encoder + closed-loop driver included; feedback/correction; documented kit | Lower torque margin; may not fit final belt/load sizing |
| 2 NEMA 23 closed-loop 3.0 N·m + CL57Y-V20 | $212.44 | Much higher holding torque; encoder + driver included; priced close to #1 | Larger, heavier and possibly excessive for printed frame |
| 3 NEMA 17 open-loop + encoder 0.52 N·m + DM542T | $153.60* | Lowest listed base cost; direct encoder data can detect motor-position mismatch | Extra electronics and STM32 code; no automatic correction |

*Option 3 price includes four encoder-equipped motors and four open-loop drivers, but excludes encoder line receivers, wiring, power, and controller development. Options 1–2 are motor + encoder + closed-loop driver kits; all listed prices exclude tax, shipping, power supplies, pulley hardware and limit switches.

## Our preliminary pick

**Option 1 — NEMA 17 closed-loop.** It best matches a compact printed prototype focused on accurate movement. If torque calculations or bench tests show insufficient margin, consider the NEMA 23 closed-loop kit. For a price-conscious team that wants to implement its own step-error monitoring, Option 3 is a learning-oriented alternative, not necessarily the simplest route.

## Key technical points

- Motor encoders detect motor-shaft errors, not necessarily belt slip or actual carriage position.
- These drivers use STEP/DIR for motion (RS232 for setup/debugging). Four motor UARTs may not be needed; verify STM32 signal interfacing.
- Synchronize both motors on each axis. Validate the printed structure and independent emergency-stop hardware before any suspended-load tests.

## Prices and product links

1. NEMA 17 closed-loop, kit 1-CL42T-S05-V41 — $48.86 each; 52 Ncm, 1000 PPR encoder; CL42T 24–48 V

2. NEMA 23 closed-loop, kit 1-CL57Y-S30-V20 — $53.11 each; 3.0 N·m, 1000 PPR encoder; CL57Y 24–50 V

3A. Encoder-equipped NEMA 17 motor 17HS19-2004D-E1K — $18.75 each

3B. Open-loop driver DM542T — $19.65 each; 18–50 V, STEP/DIR; no encoder control

*Price check: November 3, 2026. Listings and stock can change. The motor/driver product pages include technical documentation and wiring references.*